

Executive Summary& Technical Report:

Hydropower Analytics Dashboard

Project Overview

This report documents the end-to-end development of the **Hydropower Analytics Dashboard** built using Power BI. The objective of this dashboard is to analyze energy efficiency, water usage patterns, cooling technology distributions, and environmental water risk classifications across a global sample of **3,000 facilities** spanning multi-year operations (2019–2025).

1. Data Cleaning & Transformation (Power Query)

Before visual layout and dashboard design, the raw dataset underwent systematic data cleaning and transformation in Power Query Editor:

- **Null & Duplicate Auditing:** Validated facility energy and water metrics to remove missing values and invalid records across geographic regions and operator classifications.
- **Data Type Enforcement:** Explicitly assigned appropriate data types across key variables:
 - **Efficiency & Usage:** Decimal / Whole Number (PUE, Total kWh, Total Water Gallons)
 - **Categorical Fields:** Text (Country, Owner Company, Cooling Technology, Water Stress Tier)
 - **Temporal Attributes:** Date / Integer (Year)

Categorical Standardizations: Standardized water stress risk classifications (**Medium, High, Low**) and cooling methods (**Air Cooled, Evaporative, Liquid Cooled**).

2. Data Modelling & DAX Measures

Custom DAX (Data Analysis Expressions) measures were created to drive key metrics and interactive visuals:

- **Average Power Usage Effectiveness (PUE)**

Avg PUE = AVERAGE(Data_Centre_Hybrid[PUE])

- **Total Facility Count**

Total Facilities = DISTINCTCOUNT(Data_Centre_Hybrid[Facility_ID])

- **Total Energy Consumption (kWh)**

Total kWh = SUM(Data_Centre_Hybrid[WUE_L_per_kWh])

- **Total Water Consumption (Gallons)**

Total Water Gallons =
SUM(Data_Centre_Hybrid[Daily_Water_Usage_Gallons])

3. Key Analytical Insights

- **Operational Baselines:** Across **3,000 tracked facilities**, the average Power Usage Effectiveness (PUE) sits at **1.66**, generating an aggregate energy demand of **15.25K kWh** and total water utilization of **1.46 billion gallons**.
- **Operator Energy Share:** **Digital Realty Trust** leads overall energy consumption, followed by **Equinix**, **CenturyLink**, **Cyxtera**, and **Zayo Group LLC**.
- **Water Consumption Dynamics:** Annual water consumption shows a steady upward trajectory from 2019 to 2025, expanding from **~180M gallons to over 230M gallons**.
- **Cooling Technology Distribution:** **Air Cooled** technologies dominate energy utilization at **495.00K kWh**, followed by **Evaporative** systems at **430.26K kWh**, while **Liquid Cooled** systems account for **43.87K kWh**.

- **Water Risk Exposure:** The majority of facilities are located in areas classified as **Medium** and **High** water stress tiers, indicating significant operational reliance on vulnerable water supplies.

4. Strategic Recommendations

- **Cooling System Modernization:** Accelerate the transition from traditional Air-Cooled systems to high-efficiency Liquid-Cooled infrastructure to lower total kWh demand.
- **Water Stress Mitigation:** Prioritize water recycling and efficiency initiatives for facilities located in High and Medium water-stressed regions.
- **Operator Efficiency Benchmarking:** Work with major operators like Digital Realty Trust and Equinix to establish PUE reduction targets across their facility footprint.